

Finite σ -Equivariant Row-Graded Magmas on $\text{AG}(2, 3)$: Landscape, Intrinsic Selection, and Linearization

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Abstract

We study the finite landscape of σ -equivariant row-graded magmas on $M = S \times S$, $S = \mathbb{F}_3$, equivalently the point set of $\text{AG}(2, 3)$, with left-row output and a fixed same-row off-diagonal Steiner rule. The landscape has normal form

$$\Omega' \cong S^{18} \times S^3 \cong S^{21}, \quad |\Omega'| = 3^{21} = 10\,460\,353\,203.$$

Two finite atlas theorems first map the ambient space. The associativity atlas satisfies

$$63 \leq \text{Assoc} \leq 597,$$

with exact endpoint loci and partition function. The hidden-continuation frontier satisfies

$$\min H_{tot} = -2268, \quad \max H_{tot} = 7302, \quad \max_{N_-=0} H_{tot} = 7020.$$

Inside this landscape, four internal finite predicates

$$\mathcal{H}_{acc} = 0, \quad H_{diag} = 0, \quad (r, r)^2 = (r, r), \quad \text{AbsTrans} = \{C, J\}$$

select a unique point, the right-row projection magma RP . After selection, RP has a finite linearization passport: rank-three left translations, a 51-element positive composition envelope, an endomorphism and intertwiner classification, a binary term algebra of size 630, a row-contraction matrix bridge, a hidden-continuation operator identity, carrier-level compact companions, and a signed symplectic lift of the pure C/J geometry.

The theorem package is organized as

$$\text{finite landscape} \implies \text{intrinsic finite selection} \implies \text{linearization of RP}.$$

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Introduction and main theorem

Object

Let

$$S = \mathbb{F}_3, \quad M = S \times S.$$

The first coordinate is the row and the second coordinate is the column. The paper studies binary operations on M satisfying left-row output, the fixed same-row off-diagonal Steiner rule, and equivariance under the simultaneous shift $\sigma(r, c) = (r + 1, c + 1)$. The normal coordinates are

$$(A, B, d) \in S^9 \times S^9 \times S^3,$$

where A and B encode the two ordered cross-row representatives and d encodes the same-point diagonal. The resulting finite landscape is

$$\Omega' \cong S^{18} \times S^3 \cong S^{21}.$$

Main theorem package

Main theorem package. Let Ω' be the row-graded landscape on $M = S \times S$ satisfying left-row output, the fixed same-row off-diagonal Steiner rule, and σ -equivariance. Then:

1. The landscape has normal form

$$\Omega' \cong S^{18} \times S^3 \cong S^{21}, \quad |\Omega'| = 3^{21} = 10\,460\,353\,203.$$

2. The associativity atlas satisfies

$$63 \leq \text{Assoc} \leq 597,$$

with exact endpoint loci and partition function. The hidden-continuation frontier satisfies

$$\min H_{tot} = -2268, \quad \max H_{tot} = 7302, \quad \max_{N_-=0} H_{tot} = 7020.$$

3. The four internal selector predicates

$$\mathcal{H}_{acc} = 0, \quad H_{diag} = 0, \quad (r, r)^2 = (r, r), \quad \text{AbsTrans} = \{C, J\}$$

select the right-row projection magma RP uniquely, with multiplication

$$(r_1, c_1) \cdot (r_2, c_2) = \begin{cases} (r_1, r_2), & r_1 \neq r_2, \\ (r_1, \overline{c_1 c_2}), & r_1 = r_2, \ c_1 \neq c_2, \\ (r_1, r_1), & r_1 = r_2, \ c_1 = c_2. \end{cases}$$

4. The selected magma has the post-selection passport proved in Part III: left translations and positive composition envelope, endomorphisms and intertwiners, binary term functions, matrix row-contraction, a hidden-continuation operator bridge, carrier-level compact companions, and a signed symplectic lift.
5. The finite atlas and passport claims are supported by the certificate protocol recorded in the companion material.

Proof. The normal form is Theorem 1.1. The associativity atlas is Theorem 3.5, with endpoint and partition data in Theorems 3.6 and 3.7. The hidden-continuation frontier is Theorem 4.2. The selection statement is Theorem 8.1. The post-selection passport is given by Lemma 9.1, Theorems 10.2, 10.3, 10.5, 10.7, 11.2, and 12.3, together with Propositions 10.6, 11.3, 11.4, and 13.1, and Theorems 14.1, 15.1, and 16.1. The certificate interface is recorded in Section A. $\square$

Proof architecture

The proof separates the roles of the invariants and predicates.

component	role
Part I: finite landscape	normal form, effective symmetry, associativity atlas, and hidden-continuation frontier
Part II: intrinsic selection	four selector predicates and the uniqueness proof for RP
Part III: linearization	algebraic and carrier-level structures exposed after RP is fixed
Companion material	finite certificate protocol, RP one-page passport, structural reading, open bridges, and assistance note

Associativity and the $\mathcal{H}$ -frontier are atlas invariants. The four conditions in the main theorem package are selector predicates. The compact Lie groups appearing in Part III are carrier-level compact companions attached to specified linear structures on the shared carrier $k[M] \cong \text{Mat}_3(k)$.

Guide to the paper

Part I constructs the finite landscape and records its two global atlases. Part II applies the internal selector and proves uniqueness of RP. Part III fixes the selected multiplication and develops its operator, term-function, endomorphism, matrix, compact-companion, and symplectic structures. The companion material records the certificate path, a compact passport for the selected point RP, the row-complement comparison, the Landauer/Shannon structural reading, and open bridge problems.

Part I

The finite landscape

Part I constructs the ambient finite space and records the two global atlases used later: associativity and hidden continuation. The statements here describe the landscape before any selector is applied.

1 Carrier, axioms, and normal form

1.1 Carrier and shift

Let

$$S = \mathbb{F}_3, \quad M = S \times S.$$

All arithmetic in row and column coordinates is in S . Equivalently, $M = \mathbb{F}_3^2$ is the point set of the affine plane $\text{AG}(2, 3)$, with the chosen row-column coordinates fixed throughout. For $x = (r, c) \in M$, write

$$\text{row}(x) = r, \quad \text{col}(x) = c,$$

and let

$$F_r = \{r\} \times S$$

be the row fiber. We use the cyclic shift

$$\sigma(r, c) = (r + 1, c + 1).$$

Thus σ translates rows and columns simultaneously.

The row-relative offset $c - r$ is invariant under σ . Hence M decomposes into the three σ -orbits

$$D_i = \{(r, r + i) : r \in S\}, \quad i \in S.$$

The orbit D_0 is the diagonal orbit, while D_1 and D_2 are the two non-diagonal orbits.

For distinct $i, j \in S$, write $\overline{ij}$ for the third element of $S \setminus \{i, j\}$. Equivalently, in $\mathbb{F}_3$,

$$\overline{ij} = -i - j \quad (i \neq j).$$

1.2 Axioms

The landscape Ω' consists of binary operations $\cdot$ on M satisfying the following three conditions.

AX0, left-row output.

For all $x, y \in M$,

$$\text{row}(x \cdot y) = \text{row}(x).$$

Equivalently, $x \cdot y \in F_{\text{row}(x)}$.

σ -equivariance.

For all $x, y \in M$,

$$\sigma(x \cdot y) = \sigma(x) \cdot \sigma(y).$$

AX2, same-row off-diagonal Steiner rule.

For all $r, i, j \in S$ with $i \neq j$,

$$(r, r + i) \cdot (r, r + j) = (r, r + \overline{ij}).$$

Equivalently, in absolute column notation,

$$(r, c_1) \cdot (r, c_2) = (r, \overline{c_1 c_2}) \quad (c_1 \neq c_2).$$

This is the Steiner multiplication on a three-point row fiber [23, 9].

AX0 fixes only the output row. The output column remains free in the cross-row branch and on the same-point diagonal branch.

1.3 Normal coordinates

It is convenient to use row-relative coordinates adapted to this orbit decomposition. A point of row r in D_i is written as $(r, r + i)$, where $i \in S$ is the column offset from the row. Thus i is the σ -orbit label.

A normal coordinate is a triple

$$(A, B, d) \in S^9 \times S^9 \times S^3,$$

where

$$A = (A_{ij})_{i,j \in S}, \quad B = (B_{ij})_{i,j \in S}, \quad d = (d_0, d_1, d_2).$$

The associated multiplication is defined by the four formulas

$$(r, r + i) \cdot (r + 1, r + 1 + j) = (r, r + A_{ij}),$$

$$(r, r + i) \cdot (r + 2, r + 2 + j) = (r, r + B_{ij}),$$

$$(r, r + i) \cdot (r, r + i) = (r, r + d_i),$$

and, for $i \neq j$,

$$(r, r + i) \cdot (r, r + j) = (r, r + \overline{ij}).$$

The first two arrays record the two ordered cross-row representatives $(0, 1)$ and $(0, 2)$. The vector d records the same-point branch by σ -orbit label: d_i is the output offset of x^2 for $x \in D_i$. The fourth formula is AX2.

Conversely, any operation satisfying AX0, σ -equivariance, and AX2 has a unique coordinate triple (A, B, d) : after applying a power of σ , every cross-row product reduces to one of the two row-pair representatives above, and every same-point product reduces to one of the three diagonal coordinates.

1.4 Normal form theorem

Theorem 1.1. The AX0+AX2, σ -equivariant landscape has normal form

$$\Omega' \cong S^{18} \times S^3 \cong S^{21}.$$

Consequently

$$|\Omega'| = 3^{21} = 10\,460\,353\,203.$$

Proof. AX0 fixes the output row of every product. AX2 fixes all same-row off-diagonal output columns. The same-point diagonal branch has three row-relative coordinates d_0, d_1, d_2 . For cross-row products, σ -equivariance reduces the ordered row-pair data to the two representatives $(0, 1)$ and $(0, 2)$. Each representative has 3×3 column-offset inputs, hence contributes nine independent coordinates. Thus the cross-row branch contributes 18 coordinates and the diagonal branch contributes 3 coordinates.

The coordinate construction above gives a product satisfying AX0, σ -equivariance, and AX2 for every $(A, B, d) \in S^9 \times S^9 \times S^3$, and the preceding paragraph recovers these coordinates uniquely from any such product. Hence $\Omega' \cong S^{21}$, and its cardinality is 3^{21} . $\square$

2 Effective symmetry

The oriented row-graded setup fixes the generator $s \mapsto s + 1$ through $\sigma(r, c) = (r + 1, c + 1)$. For landscape compression we use the normalizer of this cyclic shift in $\text{Sym}(S)$. Translations $s \mapsto s + \beta$ commute with the shift and act trivially on the row-relative normal coordinates. The remaining normalizer class is represented by the reflection $\nu(s) = -s$, which conjugates σ to σ^{-1} . Since σ -equivariance is equivalent to σ^{-1} -equivariance, this reflection preserves Ω' as a landscape-compression symmetry.

On normal coordinates it induces the involution

$$\iota(A, B, d) = (A', B', d')$$

with indices in S and

$$A'_{ij} = -B_{-i, -j}, \quad B'_{ij} = -A_{-i, -j}, \quad d'_i = -d_{-i}.$$

We denote the resulting residual effective action by

$$G_{\text{eff}} = \langle \iota \rangle \cong C_2.$$

On the diagonal coordinate sector $\Delta = S^3$, the same residual reflection gives the compression

$$|\Delta/G_{\text{eff}}| = \frac{3^3 + 3}{2} = 15.$$

Proposition 2.1. The effective orbit count of the full normal-coordinate landscape is

$$|\Omega'/G_{\text{eff}}| = \frac{3^{21} + 3^{10}}{2}.$$

Proof. Burnside's lemma applies to the displayed involution. The identity contributes 3^{21} . For a fixed point of ι , the nine entries of A are free and then determine B by $B_{ij} = -A_{-i,-j}$. The diagonal equations are

$$d_0 = -d_0, \quad d_1 = -d_2, \quad d_2 = -d_1,$$

so $d_0 = 0$ and d_1 is free. Hence the fixed locus has $9 + 1 = 10$ ternary degrees of freedom and size 3^{10} . Therefore

$$|\Omega'/G_{\text{eff}}| = \frac{3^{21} + 3^{10}}{2}.$$

□

The orbit count is a landscape compression. The residual reflection records the finite normal-coordinate symmetry available at the atlas level. The directed-edge geometry used later is formulated on $M^\times$ itself.

3 Associativity atlas

3.1 Block decomposition

The associativity statistic is a global atlas invariant on Ω' . For $x \in \Omega'$, define

$$\text{Assoc}(x) = \#\{(u, v, w) \in M^3 : (u \cdot v) \cdot w = u \cdot (v \cdot w)\}.$$

Since $|M| = 9$, this statistic lies between 0 and 729.

Theorem 3.1. In normal coordinates,

$$\text{Assoc} = 3 \text{rawAssoc},$$

where rawAssoc is a normalized sum over $3^5 = 243$ triples and decomposes as

$$\text{rawAssoc} = \text{rawRRR} + \text{rawRRS} + \text{rawRSR} + \text{rawSRR} + \text{rawDIST}.$$

The five blocks correspond to the row patterns

$$\text{RRR}, \quad \text{RRS}, \quad \text{RSR}, \quad \text{SRR}, \quad \text{DIST}.$$

Proof. By σ -equivariance the first row coordinate of a triple may be normalized to 0. The two remaining row variables and the three column variables give 3^5 normalized cases; the factor 3 restores the three translates. The displayed blocks are the five equality patterns of the three rows. □

3.2 Same-row block

The all-same-row block depends only on the diagonal vector d . Let

$$E(d) = \#\{(i, j) : i \neq j, d_i = d_j\}, \quad N(d) = \#\{(i, j) : i \neq j, d_i = j, d_j = i\}.$$

Lemma 3.2. The same-row contribution is

$$rawRRR(d) = 9 + E(d) + 2N(d).$$

The four diagonal classes give raw same-row values

$$9, \quad 11, \quad 13, \quad 19.$$

In particular,

$$rawRRR(000) = 19.$$

Proof. In one row, unequal column pairs use the Steiner rule and equal pairs use d . The 27 triples split into a baseline family plus the two diagonal-sensitive families counted by $E(d)$ and $N(d)$. This gives the formula and the four class values. $\square$

3.3 Column-blind profile at $d = 000$

A cross-row rule is column-blind when its output column depends only on the two row coordinates. By σ -equivariance such a rule is determined by two anchors $a = h(0, 1)$ and $b = h(0, 2)$. Let $h_{a,b}$ be the column-blind rule

$$h(r, r+1) = r + a, \quad h(r, r+2) = r + b.$$

At $d = 000$, same-point products satisfy $(r, c)^2 = (r, r)$.

Proposition 3.3. On the column-blind, $d = 000$ stratum,

$$\begin{aligned} (rawRRR, rawRRS, rawRSR, rawSRR, rawDIST)(a, b, 000) \\ = (19, 9[a = 0] + 9[b = 0], 9[a = 0] + 9[b = 0], \\ 54, 54[a = b]). \end{aligned}$$

Consequently

$$\boxed{\text{Assoc}_{000}(h_{a,b}) = 3(73 + 18[a = 0] + 18[b = 0] + 54[a = b]) .}$$

Proof. The RRR value is Lemma 3.2. In the RRS and RSR patterns, the same-row step must return the cross-row anchor; at $d = 000$ this contributes 9 exactly when the relevant anchor is 0. In SRR , the final cross-row output ignores same-row column processing, giving all 54 normalized triples. In $DIST$, the two association orders compare the two cross-row anchors, so all 54 triples survive exactly when $a = b$. $\square$

The nine column-blind rules split as follows:

class	representatives	profile	Assoc ₀₀₀
anchored trivial	(0, 0)	(19, 18, 18, 54, 54)	489
off-anchor trivial	(1, 1), (2, 2)	(19, 0, 0, 54, 54)	381

class	representatives	profile	Assoc ₀₀₀
partly anchored nondegenerate	(0, 1), (0, 2), (1, 0), (2, 0)	(19, 9, 9, 54, 0)	273
fully off-anchor nondegenerate	(1, 2), (2, 1)	(19, 0, 0, 54, 0)	219

The fourth class consists of the selected point $RP = (a, b, d) = (1, 2, 000)$ and the row-complement comparison point $RC = (a, b, d) = (2, 1, 000)$.

3.4 Global range

Lemma 3.4 (Associativity atlas certificate protocol). The global associativity atlas is used through five finite obligations.

1. **A1, block reduction.** The associativity statistic is reduced by σ -normalization to the five row-pattern blocks

$$rawAssoc = rawRRR + rawRRS + rawRSR + rawSRR + rawDIST$$

of Theorem 3.1.

2. **A2, diagonal decomposition.** The same-row block is evaluated from the diagonal vector by Lemma 3.2, giving the four possible values

$$9, \quad 11, \quad 13, \quad 19.$$

3. **A3, fixed-diagonal cross-range certificates.** For each diagonal class, the certificate verifies the exact range of

$$rawCross = rawRRS + rawRSR + rawSRR + rawDIST.$$

4. **A4, endpoint witnesses and endpoint counts.** The endpoint certificates verify the normal-coordinate witnesses, endpoint counts, and orbit representatives for $Assoc = 63$ and $Assoc = 597$.
5. **A5, distribution consistency.** The partition-function certificate gives the full coefficient table for

$$Z(q) = \sum_{x \in \Omega'} q^{Assoc(x)}$$

and checks total mass, range, moments, and mode.

Any certificate satisfying A1–A5 proves the global associativity range, endpoint loci, and partition-function claims in this atlas section. $\square$

Theorem 3.5 (Global associativity atlas). On Ω' ,

$$21 \leq rawAssoc \leq 199.$$

Equivalently,

$$63 \leq Assoc \leq 597.$$

The finite certificate is organized by diagonal class:

diagonal class	rawRRR	certified rawCross	
		range	rawAssoc consequence
permutation	9	12..180	21..189
two-valued, no fixed point	11	10..162	21..173
two-valued, with fixed point	13	12..174	25..187
constant	19	12..180	31..199

Here

$$rawCross = rawRRS + rawRSR + rawSRR + rawDIST.$$

The endpoint values are therefore $rawAssoc = 21$ and $rawAssoc = 199$, or $Assoc = 63$ and $Assoc = 597$.

Proof. The hand-readable reduction is Theorem 3.1 plus Lemma 3.2. After that reduction, the global claim is exactly the A3 cross-range obligation of Lemma 3.4. Adding the certified cross-row ranges to the four possible same-row values gives the displayed global range.

The endpoint witnesses, endpoint counts, orbit representatives, and partition-function consistency are the A4–A5 obligations of Lemma 3.4. They are recorded in the `L1/tables/` atlas files listed in `SUPPORT_INDEX.txt` and checked by `L1/scripts/verify_layer1_v3_final.py`. $\square$

Atlas landmarks. The selected point has

$$Assoc(RP) = 219,$$

while the landscape minimum is 63. On the column-blind $d = 000$ profile, the value $Assoc_{000} = 219$ is attained by the two fully off-anchor nondegenerate rules RP and RC.

3.5 Endpoint loci and partition function

3.5.1 Endpoint loci

Theorem 3.6. The associativity endpoint loci have sizes

$$|\mathcal{M}_{63}| = 24, \quad |\mathcal{M}_{63}/\text{Sym}(S)| = 12,$$

and

$$|\mathcal{M}_{597}| = 6, \quad |\mathcal{M}_{597}/\text{Sym}(S)| = 3.$$

The minimum endpoint block profiles are

$$(11, 2, 2, 2, 4) \quad \text{and} \quad (9, 2, 2, 4, 4),$$

while the maximum endpoint block profile is

$$(19, 36, 36, 54, 54).$$

Proof. This is the A4 endpoint part of Lemma 3.4. The endpoint certificates record the exact solution counts by diagonal sector, orbit representatives, and explicit witnesses in normal coordinates. The minimum occurs in the permutation and two-valued/no-fixed diagonal classes. The maximum occurs in the constant diagonal class. The block profiles are obtained by evaluating the five normalized associativity blocks on the endpoint witnesses. $\square$

3.5.2 Exact associativity partition function

Theorem 3.7. The exact polynomial

$$Z(q) = \sum_{x \in \Omega'} q^{\text{Assoc}(x)}$$

has 167 nonzero coefficients and total mass

$$Z(1) = 3^{21}.$$

Its first two moments are

$$\mathbb{E}[\text{Assoc}] = 245, \quad \text{Var}(\text{Assoc}) = \frac{27820}{27}.$$

The mode occurs at

$$\text{Assoc} = 237$$

with coefficient

$$426, 317, 976.$$

Proof. This is the A5 partition-function part of Lemma 3.4. The coefficient certificate in `L1/tables/` is the exact finite distribution of Assoc on Ω' . The total mass, range, mean, variance, and mode are computed directly from that table and checked by `L1/scripts/verify_layer1_v3_final.py`. $\square$

4 Hidden-continuation frontier

4.1 Local distance reduction

The hidden-continuation statistic is the second global atlas invariant. It is a finite path-information defect. The continuation side keeps the two-step route data, while the left-translation side compares only the collapsed endpoint operators. Thus $h_q/2$ is the signed Hamming contrast between a path-resolved comparison and an endpoint-resolved comparison. Negative local values are precisely the local cancellation failures counted by N_- . The normalization $H_{tot} = 6rawH$ is the σ -orbit normalization used by the $\mathcal{H}$ -frontier certificate.

Its local index is

$$q = (b, t, a, e, f) \in S^5.$$

For a completed normal-form point $x \in \Omega'$, let M_s denote the local table read: M_1 and M_2 are the two cross-row tables, and M_0 is the same-row diagonal/complement rule. All row and column arithmetic below is in $S = \mathbb{F}_3$. Define

$$xy = M_b(a, e), \quad yz = M_{t-b}(e - b, f - b), \quad z_R = b + yz,$$

and

$$ep_L = M_t(xy, f), \quad ep_R = M_b(a, z_R).$$

For $c \in S$, define the collapsed left-translation signature

$$L_c(i, j) = M_i(c, j) \quad ((i, j) \in S^2).$$

For $s, \alpha, z \in S$, define the stepwise continuation signature

$$C_{s, \alpha, z}(i, j) = M_s(\alpha, s + M_{i-s}(z - s, j - s)) \quad ((i, j) \in S^2).$$

The associated Hamming distances are

$$D_L(c, c') = \#\{(i, j) \in S^2 : L_c(i, j) \neq L_{c'}(i, j)\},$$

and

$$D_C(s, \alpha, z; s', \alpha', z') = \#\{(i, j) \in S^2 : C_{s, \alpha, z}(i, j) \neq C_{s', \alpha', z'}(i, j)\}.$$

Direct local definition. For each of the nine test pairs $(i, j) \in S^2$, set

$$\delta_{ij}^C(q; x) = \mathbf{1}\{C_{t, xy, f}(i, j) \neq C_{b, a, z_R}(i, j)\},$$

and

$$\delta_{ij}^L(q; x) = \mathbf{1}\{L_{ep_L}(i, j) \neq L_{ep_R}(i, j)\}.$$

The nine direct local summands are

$$\Delta_{ij}(q; x) = 2(\delta_{ij}^C(q; x) - \delta_{ij}^L(q; x)),$$

displayed as

	$j = 0$	$j = 1$	$j = 2$
$i = 0$	Δ_{00}	Δ_{01}	Δ_{02}
$i = 1$	Δ_{10}	Δ_{11}	Δ_{12}
$i = 2$	Δ_{20}	Δ_{21}	Δ_{22}

and the direct hidden-continuation contribution is

$$h_q(x) = \sum_{(i, j) \in S^2} \Delta_{ij}(q; x).$$

Theorem 4.1 (Local distance reduction). For every $x \in \Omega'$ and $q \in S^5$,

$$\boxed{h_q(x) = 2(D_C(t, xy, f; b, a, z_R) - D_L(ep_L, ep_R)).}$$

Consequently the local direct definition and the Hamming-distance formula define the same global observables $rawH$ and H_{tot} .

Proof. By definition, $D_C(t, xy, f; b, a, z_R)$ is the sum of the nine indicators $\delta_{ij}^C(q; x)$, and $D_L(ep_L, ep_R)$ is the sum of the nine indicators $\delta_{ij}^L(q; x)$. Subtracting these two Hamming distances and multiplying by 2 gives exactly the displayed sum of the nine direct local summands Δ_{ij} . The symbolic reduction is recorded by the $\mathcal{H}$ -frontier certificate handles **HF-RED** and **HF-CERT-IF**. $\square$

Then

$$rawH(x) = \sum_{q \in S^5} \frac{h_q(x)}{2}, \quad H_{tot}(x) = 6 rawH(x) = 3 \sum_{q \in S^5} h_q(x),$$

and

$$\boxed{N_-(x) = 3 \#\{q \in S^5 : h_q(x) < 0\}.}$$

Thus purity is the finite system

$$N_-(x) = 0 \iff h_q(x) \geq 0 \quad (q \in S^5).$$

For the signed-layer diagnostics, set

$$H_{pos}(x) = 3 \sum_{q \in S^5} \max\{h_q(x), 0\}, \quad H_{neg_abs}(x) = 3 \sum_{q \in S^5} \max\{-h_q(x), 0\}.$$

Then

$$H_{tot}(x) = H_{pos}(x) - H_{neg_abs}(x).$$

Also write

$$h_{loc, \min}(x) = \min_{q \in S^5} h_q(x), \quad h_{loc, \max}(x) = \max_{q \in S^5} h_q(x).$$

4.2 Global frontier theorem

The global $\mathcal{H}$ -frontier theorem is supported by four finite certificate obligations.

handle	obligation	result
H1	unrestricted full-landscape range	$rawH \in [-378, 1217]$, endpoint counts 8, 12
H2	pure-frontier exclusion	$(h_q \geq 0 \ \forall q) \wedge rawH \geq 1171$ is UNSAT
H3	pure-frontier witnesses	RP and RC have $rawH = 1170, N_- = 0$
H4	signed-cancellation spike	exactly twelve points above the pure frontier, all with $(rawH, N_-) = (1217, 3)$

Theorem 4.2 (Global frontier of $\mathcal{H}$). On $\Omega' = S^{21}$:

1. The unrestricted range is

$$\boxed{\min H_{tot} = -2268, \quad \max H_{tot} = 7302.}$$

Equivalently, $rawH \in [-378, 1217]$. The endpoint multiplicities are 8 and 12.

2. The pure frontier is

$$\boxed{\max\{H_{tot} : N_- = 0\} = 7020.}$$

3. The region above the pure frontier is exactly the unrestricted maximum layer:

$$\boxed{\{H_{tot} > 7020\} = \{H_{tot} = 7302\}.}$$

This layer has 12 points, all with

$$rawH = 1217, \quad H_{tot} = 7302, \quad N_- = 3,$$

$$H_{pos} = 7308, \quad H_{neg_abs} = 6, \quad h_{loc,min} = -2, \quad h_{loc,max} = 18.$$

Proof. The analytic reduction is Theorem 4.1. H1 covers all 3^{21} normal-form words and gives the unrestricted range. For the pure frontier, $N_- = 0$ is the system $h_q \geq 0$ for all $q \in S^5$, and $H_{tot} > 7020$ is equivalent to $rawH \geq 1171$. H2 proves

$$(h_q \geq 0 \ \forall q \in S^5) \wedge rawH \geq 1171$$

UNSAT on all 18540 depth-9 live nodes:

$$18540/18540 \text{ UNSAT}, \quad \text{SAT} = 0, \quad \text{UNKNOWN} = 0.$$

H3 supplies pure witnesses at $rawH = 1170$, and H4 classifies the full layer above the pure threshold as the twelve impure maximizers. $\square$

4.3 Pure frontier and signed-cancellation layer

Corollary 4.3. RP and RC satisfy

$$H_{tot} = 7020, \quad rawH = 1170, \quad N_- = 0.$$

Thus both lie on the global pure $\mathcal{H}$ -frontier.

Proof. Direct evaluation of

$$RP = 1111111122222222000, \quad RC = 2222222211111111000$$

gives the displayed values, and Theorem 4.2 proves that no pure point has larger H_{tot} . □

Part II

Intrinsic finite selection

Part I gives the finite atlas data for the ambient landscape. Part II defines the selector and proves that its unique survivor is RP. The first two predicates are compression predicates, the third fixes the compressed diagonal convention, and the fourth is the directed-edge geometry predicate.

5 Selector predicates

For a normal-form point of Ω' , write g for its cross-row output column rule and $d = (d_0, d_1, d_2)$ for its same-point diagonal. The selector is the conjunction

$$\boxed{\mathcal{H}_{acc}(g) = 0, \quad H_{diag}(d) = 0, \quad (r, r)^2 = (r, r) \ (r \in S), \quad \text{AbsTrans} = \{C, J\}.$$

The first predicate is hidden-column access zero. The second is diagonal Shannon entropy zero. The third fixes the row-anchor representative after entropy compression; equivalently, it anchors idempotence on the diagonal orbit D_0 . The fourth chooses the intrinsic head-row continuation/reversal geometry on the six directed edges.

The first two predicates are information-theoretic: $\mathcal{H}_{acc}$ is conditional mutual information and H_{diag} is Shannon entropy [26, 10]. The Landauer–Bennett reading is the implementation-independent economy reading of avoidable hidden column access and discarded diagonal information [19, 3, 4].

6 Access-zero and diagonal compression

6.1 Column-blind cross-row rules

For $r_1 \neq r_2$, write the cross-row branch as

$$(r_1, c_1) \cdot (r_2, c_2) = (r_1, g(r_1, c_1, r_2, c_2)).$$

A cross-row rule is *column-blind* if the output column depends only on its ordered row pair:

$$g(r_1, c_1, r_2, c_2) = h(r_1, r_2).$$

Equivalently, after (r_1, r_2) is fixed, the output column is constant over the nine choices of (c_1, c_2) .

By σ -equivariance, a column-blind rule is determined by two anchors

$$a = h(0, 1), \quad b = h(0, 2),$$

and the remaining ordered row pairs satisfy

$$h(r, r + 1) = r + a, \quad h(r, r + 2) = r + b.$$

There are $3^2 = 9$ column-blind cross-row rules; denote them by $h_{a,b}$.

6.2 Hidden-column access

Let

$$\Sigma_{cross} = \{(r_1, c_1, r_2, c_2) \in S^4 : r_1 \neq r_2\}$$

with the uniform distribution, and let

$$C_{out} = g(R_1, C_1, R_2, C_2)$$

be the output-column random variable. Define

$$\mathcal{H}_{acc}(g) = I(C_{out}; C_1, C_2 \mid R_1, R_2).$$

Logarithms are base 2.

Theorem 6.1 (Access-zero theorem). For a cross-row rule g ,

$$\mathcal{H}_{acc}(g) = 0 \iff g(r_1, c_1, r_2, c_2) = h(r_1, r_2).$$

Thus the access-zero locus consists of the nine column-blind rules $h_{a,b}$.

Proof. Conditional mutual information vanishes exactly when

$$C_{out} \perp (C_1, C_2) \mid (R_1, R_2).$$

Since C_{out} is a deterministic function of (R_1, C_1, R_2, C_2) , this conditional independence is equivalent to constancy in (C_1, C_2) for every fixed ordered row pair. This is exactly column-blindness. The ordered row-pair representatives $(0, 1)$ and $(0, 2)$ then supply the two anchors $(a, b) \in S^2$. $\square$

6.3 Diagonal entropy

For a diagonal triple $d = (d_0, d_1, d_2) \in S^3$, let

$$p_s = \frac{1}{3} \# \{i : d_i = s\}.$$

The diagonal Shannon observable is

$$H_{diag}(d) = - \sum_{s \in S} p_s \log_2 p_s.$$

The possible levels are

diagonal type	multiplicities	H_{diag}
constant	$(3, 0, 0)$	0
two-valued	$(2, 1, 0)$	$\log_2 3 - \frac{2}{3}$
permutation	$(1, 1, 1)$	$\log_2 3$

Theorem 6.2 (Diagonal compression). For diagonal triples,

$$H_{diag}(d) = 0 \iff d_0 = d_1 = d_2.$$

Thus diagonal compression reduces the diagonal sector from 27 choices to

$$000, \quad 111, \quad 222.$$

Together with access-zero, the selector domain becomes 9×3 .

Proof. Entropy of a finite distribution vanishes exactly when the distribution is concentrated on one symbol. For the empirical distribution of (d_0, d_1, d_2) , this is precisely constancy of the diagonal triple. $\square$

6.4 Row-anchor idempotence

The row anchors are the points of the diagonal orbit D_0 .

Proposition 6.3 (Row-anchor idempotence). After diagonal compression, the condition

$$(r, r) \cdot (r, r) = (r, r) \quad (r \in S)$$

selects the unique constant diagonal

$$d = 000.$$

For this diagonal,

$$(r, r + i)^2 = (r, r) \quad (r, i \in S),$$

so the idempotent points are exactly the diagonal orbit D_0 .

Proof. On the compressed diagonal sector $d = sss$, the equation $(r, r)^2 = (r, r)$ gives $s = 0$. Hence the unique compressed diagonal compatible with row-anchor idempotence is 000. The displayed square formula follows by substituting $d = 000$ into the same-point branch. $\square$

The first three selector predicates therefore reduce the normal-form coordinates to

$$h_{a,b}, \quad (a, b) \in S^2, \quad d = 000.$$

7 Directed-edge geometry

7.1 The maps C and J

Let

$$M^\times = \{(r, c) \in M : r \neq c\}$$

be the six-point directed-edge set of the triangle on vertex set S . In the orbit notation above,

$$M^\times = D_1 \sqcup D_2.$$

The point (r, c) is the directed edge with tail r and head c . Define

$$C(r, c) = (c, \overline{rc}), \quad J(r, c) = (c, r).$$

The map C advances the edge through the third vertex, while J reverses it.

Theorem 7.1 (Directed-edge dihedral relations). On $M^\times$,

$$C^3 = 1, \quad J^2 = 1, \quad JCJ = C^{-1}.$$

Thus C and J generate the dihedral geometry of the directed triangle.

Proof. Applying C three times gives

$$(r, c) \mapsto (c, \overline{rc}) \mapsto (\overline{rc}, r) \mapsto (r, c).$$

Applying J twice reverses the edge twice. Finally,

$$JCJ(r, c) = JC(c, r) = J(r, \overline{rc}) = (\overline{rc}, r) = C^{-1}(r, c).$$

$\square$

For every relabelling $\rho \in \text{Sym}(S)$, acting by $\rho_\times(r, c) = (\rho(r), \rho(c))$, one has

$$\rho_\times C = C \rho_\times, \quad \rho_\times J = J \rho_\times.$$

The pair $\{C, J\}$ is the intrinsic head-row pair:

$$\{C(r, c), J(r, c)\} = \{y \in M^\times : \text{tail}(y) = c\}.$$

7.2 Absorption transforms

For $x \in M^\times$, define

$$\text{Abs}(x) = \{y \in M^\times \setminus \{x\} : x \cdot y = x\}.$$

On the anchored selector domain $h_{a,b}$, $d = 000$, absorption of $x = (r, c)$ is governed by the cross-row equation

$$h(r, r_y) = c.$$

The pure C/J predicate is the finite condition

$$\text{Abs}(x) = \{C(x), J(x)\} \quad (x \in M^\times).$$

Equivalently, the anchored rule has two absorbed-neighbour maps, and they form

$$\text{AbsTrans} = \{C, J\}.$$

7.3 Pure C/J

Theorem 7.2 (Pure C/J selector on the anchored domain). For a column-blind rule $h_{a,b}$ with $d = 000$,

$$\text{AbsTrans} = \{C, J\} \iff (a, b) = (1, 2).$$

The corresponding cross-row branch is the right-row projection branch.

Proof. Let $x = (r, c) \in M^\times$, and write $\delta = c - r \in \{1, 2\}$. For the two cross rows of a potential absorber, the absorption equation becomes

$$r_y = r + 1 \Rightarrow a = \delta, \quad r_y = r + 2 \Rightarrow b = \delta.$$

The pure C/J predicate requires the absorbed neighbours of (r, c) to be exactly

$$J(r, c) = (c, r), \quad C(r, c) = (c, \bar{r}c),$$

so both absorbed neighbours must lie in row c .

For $\delta = 1$, this forces absorption in row $r + 1 = c$, hence $a = 1$, and excludes absorption in row $r + 2$, hence $b \neq 1$. For $\delta = 2$, it forces absorption in row $r + 2 = c$, hence $b = 2$, and excludes absorption in row $r + 1$, hence $a \neq 2$. Therefore $(a, b) = (1, 2)$.

Conversely, if $(a, b) = (1, 2)$, then for every $x = (r, c)$ the only absorbing cross row is row c , and the two off-diagonal points in that row are (c, r) and $(c, \bar{r}c)$, namely $J(x)$ and $C(x)$. Thus $\text{AbsTrans} = \{C, J\}$. $\square$

8 Uniqueness theorem

Theorem 8.1 (Uniqueness theorem). The four conditions

$$\mathcal{H}_{acc} = 0, \quad H_{diag} = 0, \quad (r, r)^2 = (r, r) \ (r \in S), \quad \text{AbsTrans} = \{C, J\}$$

select RP uniquely inside Ω' .

Proof. Access-zero gives a column-blind rule $h_{a,b}$ by Theorem 6.1. Diagonal entropy zero gives a constant diagonal by Theorem 6.2, and row-anchor idempotence fixes it to $d = 000$ by Proposition 6.3. On this anchored domain, the pure C/J selector is equivalent to $(a, b) = (1, 2)$ by Theorem 7.2. This is exactly the right-row projection multiplication, with

$$(r_1, c_1) \cdot (r_2, c_2) = (r_1, r_2) \quad (r_1 \neq r_2),$$

and same-point diagonal $d = 000$. Hence the survivor is unique and is RP. $\square$

Part III

Linearization of the selected magma

Part III fixes the selected multiplication RP and records the structures that appear after this choice. The first layer is algebraic: row ownership, left translations, composition envelope, row-fiber algebra, endomorphisms, intertwiners, and binary term functions. The second layer is carrier-level: row-contraction on $k[M] \cong \text{Mat}_3(k)$, the hidden-continuation operator bridge, compact companions, and the signed symplectic lift of the pure directed-edge geometry.

9 The selected operation

9.1 Multiplication and ground-field conventions

The selected multiplication is

$$(r_1, c_1) \cdot (r_2, c_2) = \begin{cases} (r_1, r_2), & r_1 \neq r_2, \\ (r_1, \overline{c_1 c_2}), & r_1 = r_2, \ c_1 \neq c_2, \\ (r_1, r_1), & r_1 = r_2, \ c_1 = c_2. \end{cases}$$

The first case is the right-row projection branch; the second is the fixed same-row Steiner branch; the third is diagonal anchoring.

The finite magma statements are set-theoretic. When the carrier is linearized, let k be a ground field and let $k[M]$ be the vector space with basis e_x , $x \in M$. Rank and image statements for individual finite maps are valid over any ground field. Linear span statements are over the chosen k . Spectral, Jordan, and fiber-algebra statements involving $t \pm 1$ or idempotents with coefficients $1/2$ are stated under

$$\text{char } k \neq 2.$$

The carrier-level compact-companion and signed symplectic bridge statements are over $\mathbb{R}$ unless stated otherwise.

9.2 Row-owner law

For a nonempty term $t(x_1, \dots, x_n)$, define its **owner** $\text{ow}(t)$ to be the leftmost variable appearing in the term tree.

Lemma 9.1 (Row-owner lemma). For every RP term $t(x_1, \dots, x_n)$,

$$\text{row}(t(x_1, \dots, x_n)) = \text{row}(\text{ow}(t)).$$

Proof. The assertion is immediate for a variable. If $t = u \cdot v$, then the left-row output axiom gives $\text{row}(t) = \text{row}(u \cdot v) = \text{row}(u)$. The induction hypothesis identifies this row with the row of the leftmost variable of u , which is also the leftmost variable of $u \cdot v$. $\square$

The row-owner lemma is the common source of the row-target product law for left-translation compositions and the owner coordinate in the binary term algebra.

10 Left translations and operator envelope

For $x \in M$, write $L_x(y) = x \cdot y$.

10.1 Rank and image

Proposition 10.1. For $x = (r, c)$,

$$L_{(r,c)}(s, d) = \begin{cases} (r, s), & s \neq r, \\ (r, r), & s = r, d = c, \\ (r, \overline{cd}), & s = r, d \neq c. \end{cases}$$

Consequently $L_{(r,c)}(M) = F_r$.

Proof. The three cases are exactly the cross-row projection, diagonal anchoring, and same-row Steiner branches of RP. They produce all three points of F_r . $\square$

Theorem 10.2. For every $x \in M$,

$$\text{rank } L_x = 3, \quad \text{Im } L_x = F_{\text{row}(x)}.$$

Moreover,

$$\dim \text{span}\{L_x : x \in M\} = 9.$$

Proof. Proposition 10.1 shows that each L_x maps onto the three-point row fiber $F_{\text{row}(x)}$, hence has rank three after linearization. Operators with different target rows have disjoint codomain support. For a fixed row, the three restrictions to the row fiber are linearly independent; in normalized row 0 they contain the matrices

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus each row contributes dimension 3, and the total dimension is 9. The L3/ operator rank and spectrum audit tables record the nine-row calculation. $\square$

Assume $\text{char } k \neq 2$.

Theorem 10.3. If $x = (r, r)$, then

$$\mu_{L_x}(t) = t(t-1)(t+1)$$

and L_x is semisimple. If $x = (r, c)$ with $r \neq c$, then

$$\mu_{L_x}(t) = t^2(t-1)(t+1),$$

with zero-primary Jordan structure $J_2(0) \oplus J_1(0)^{\oplus 5}$.

Proof. Every $L_{(r,c)}$ maps $k[M]$ onto kF_r , so the nonzero spectral part is the restriction to kF_r . For $c = r$, this restriction fixes (r, r) and swaps the two off-diagonal basis vectors. For $c \neq r$, the restriction has eigenvalues $0, 1, -1$, and its one-dimensional kernel lies inside the image, producing exactly one length-two zero block. The operator-spectrum table audits the calculation for all nine translations. $\square$

10.2 Positive composition envelope

Let $\text{Mon}^+\langle L_x \rangle$ be the positive composition envelope generated by the nine left translations. The superscript $+$ means nonempty words in the generators; the identity map is not included. It is a semigroup under composition, and the unital monoid obtained by adjoining the identity has one additional element. For $r \in S$, let $\mathcal{W}_r$ be the component of maps whose image lies in F_r .

For fixed r , define

$$\tau_{r,c}(d) = \begin{cases} r, & d = c, \\ \overline{cd}, & d \neq c, \end{cases} \quad \mathcal{P}_r = \langle \tau_{r,c} : c \in S \rangle.$$

In normalized row 0,

$$\mathcal{P}_0 = \{012, 021, 200, 100, 011, 022\}.$$

For $\psi \in \mathcal{P}_r$ and $c \in S$, set

$$N_{r,\psi,c}(s, d) = \begin{cases} (r, \psi(\tau_{r,c}(d))), & s = r, \\ (r, \psi(s)), & s \neq r, \end{cases}$$

and let $K_{r,t}$ be the constant map to (r, t) .

Proposition 10.4 (Fixed-row composition normal form). For every $r \in S$,

$$\mathcal{W}_r = \{N_{r,\psi,c} : \psi \in \mathcal{P}_r, c \in S\} \cup \{K_{r,t} : t \in S\}.$$

There are 14 nonconstant maps of the first type and 3 constants; hence $|\mathcal{W}_r| = 17$.

Proof. A positive composition is owned by its leftmost generator. If all later generators have the same target row r , the word has the displayed $N_{r,\psi,c}$ form. If some later generator has target row different from r , the leftmost row- r generator sees a cross-row input and the composition becomes constant. The normalized finite count $18 - 4 = 14$ for the nonconstant forms is audited in the composition envelope table. $\square$

Theorem 10.5. The positive composition envelope generated by the left translations has size

$$|\text{Mon}^+\langle L_x \rangle| = 51.$$

The identity map is not contained in it; adjoining the identity gives the unital generated monoid of size 52. More precisely,

$$\text{Mon}^+\langle L_x \rangle = \mathcal{W}_0 \sqcup \mathcal{W}_1 \sqcup \mathcal{W}_2, \quad |\mathcal{W}_r| = 17.$$

Its linear envelope decomposes as

$$\mathcal{E} = W_0 \oplus W_1 \oplus W_2, \quad \dim W_r = 9, \quad \dim \mathcal{E} = 27,$$

and satisfies $W_r W_s \subseteq W_r$.

Proof. Proposition 10.4 gives $|\mathcal{W}_r| = 17$, and the three target-row components are disjoint. Every positive word has image contained in a single row fiber, so the identity map on M is not a positive word. The row-owner law gives $W_r W_s \subseteq W_r$. The linear dimension claim is the row-wise basis-reduction audit recorded in the L3/ composition-envelope and product-law tables. $\square$

10.3 Row-fiber algebra

Assume $\text{char } k \neq 2$. For each row r , define

$$A_{5,r} = \langle L_{(r,c)}|_{kF_r} : c \in S \rangle \subseteq \text{End}(kF_r).$$

Proposition 10.6. The algebra $A_{5,r}$ has dimension 5. Its center is kI , its radical is two-dimensional and square-zero, and

$$A_{5,r} / \text{Rad}(A_{5,r}) \cong k^3.$$

For $r = 0$, one basis is

$$L_{(0,0)}|_{kF_0}, \quad L_{(0,1)}|_{kF_0}, \quad L_{(0,2)}|_{kF_0}, \quad I, \quad (L_{(0,1)}|_{kF_0})^2.$$

Proof. By σ -equivariance it is enough to work in row 0. The five-dimensional multiplication table, center, radical, quotient, and operator-boundary checks are recorded in the auxiliary fiber-algebra certificate files listed in `SUPPORT_INDEX.txt`. $\square$

Theorem 10.7. On the six directed edges,

$$\mathbb{R}[M^\times] \cong \mathbf{1} \oplus \text{sign} \oplus 2 \text{std}.$$

For RP, the absorption matrix is

$$A_{abs} = (P_C + P_J)^\top$$

and has spectrum

$$\{2, 0, 0, 0, -1, -1\}.$$

Proof. The six directed edges carry the permutation representation of the directed triangle. The pure absorption transitions of RP are exactly C and J , so the absorption matrix is $(P_C + P_J)^\top$. The transition and spectrum tables audit the finite calculation. $\square$

11 Endomorphisms and intertwiners

For a map $\rho : S \rightarrow S$, write $\varphi_\rho(r, c) = (\rho(r), \rho(c))$. For $t \in S$, write $e_t(r, c) = (t, t)$.

For a row r , set

$$c \star_r d = \begin{cases} r, & c = d, \\ \overline{cd}, & c \neq d. \end{cases}$$

Lemma 11.1 (Row-fiber homomorphisms). If $\phi : S \rightarrow S$ satisfies $\phi(r) = t$ and

$$\phi(c \star_r d) = \phi(c) \star_t \phi(d) \quad (c, d \in S),$$

then ϕ is either the constant map to t , or a permutation of S sending r to t .

Proof. After translating to $r = t = 0$, write $\phi(1) = u$, $\phi(2) = v$. The relations $0 \star_0 1 = 2$ and $0 \star_0 2 = 1$ force

$$(u, v) = (0, 0), \quad (1, 2), \quad (2, 1),$$

which are respectively the constant solution and the two permutations fixing 0. $\square$

Theorem 11.2. The endomorphism monoid of the selected RP magma is

$$\text{End}(M, \cdot) = \{\varphi_\rho : \rho \in \text{Sym}(S)\} \cup \{e_0, e_1, e_2\}.$$

Proof. The idempotents of RP are exactly the diagonal points (r, r) . Thus an endomorphism sends row fibers to row fibers over a map $\rho : S \rightarrow S$, and its restriction to each row fiber is governed by Lemma 11.1. The cross-row rule forces all nonconstant row restrictions to be the same permutation ρ . If ρ fails to be injective, the cross-row equations force all affected row restrictions to collapse, and then the third row collapses as well; the map is one of the e_t . Conversely, simultaneous permutations and diagonal collapses preserve the three branches of the RP multiplication. $\square$

For an endomorphism φ , let Hom_φ be the space of intertwiners satisfying

$$TL_x = L_{\varphi(x)}T \quad (x \in M).$$

Let $V = k[M]$, and let

$$\varepsilon : V \rightarrow k, \quad \varepsilon \left(\sum_{m \in M} a_m e_m \right) = \sum_{m \in M} a_m$$

be the total augmentation functional. For $r \in S$, set

$$d_r = e_{(r,r)}, \quad o_r = \sum_{c \neq r} e_{(r,c)}, \quad U_r = \text{span}\{d_r, o_r\}.$$

Here $u\varepsilon$ denotes the rank-one operator $v \mapsto \varepsilon(v)u$.

Proposition 11.3 (Constant intertwiners). For each diagonal collapse $e_r(x) = (r, r)$,

$$\text{Hom}_{e_r} = \{u\varepsilon : u \in U_r\} = \left\{ \left(\alpha e_{(r,r)} + \beta \sum_{c \neq r} e_{(r,c)} \right) \varepsilon : \alpha, \beta \in k \right\}.$$

In particular, $\dim \text{Hom}_{e_r} = 2$.

Proof. Each L_x is induced by a map $M \rightarrow M$, hence every column of its matrix has sum one and $\varepsilon L_x = \varepsilon$. If $u \in \text{Fix}(L_{(r,r)})$, then

$$(u\varepsilon)L_x = u\varepsilon \quad \text{and} \quad L_{(r,r)}(u\varepsilon) = (L_{(r,r)}u)\varepsilon = u\varepsilon,$$

so $u\varepsilon \in \text{Hom}_{e_r}$.

Conversely, if $T \in \text{Hom}_{e_r}$, then

$$TL_x = L_{(r,r)}T = TL_y \quad (x, y \in M),$$

so T kills

$$\mathcal{D} = \text{span}\{(L_x - L_y)v : x, y \in M, v \in V\}.$$

Because $\varepsilon L_x = \varepsilon$, one has $\mathcal{D} \subseteq \ker \varepsilon$. The reverse inclusion is finite: the graph generated by relations $x \cdot z \sim y \cdot z$, with common source z , is connected. The Front-C verifier records an explicit eight-edge spanning tree, so the differences in $\mathcal{D}$ span the augmentation-zero hyperplane. Thus $\mathcal{D} = \ker \varepsilon$, and T factors uniquely as $T = u\varepsilon$.

Substitution into the intertwining equation gives $L_{(r,r)}u = u$. Finally, $L_{(r,r)}$ has image in the row fiber $F_r = \text{span}\{e_{(r,c)} : c \in S\}$; on this fiber it fixes $e_{(r,r)}$ and swaps the two off-diagonal basis vectors. Therefore

$$\text{Fix}(L_{(r,r)}) = \text{span} \left\{ e_{(r,r)}, \sum_{c \neq r} e_{(r,c)} \right\} = U_r,$$

which proves the formula. $\square$

Proposition 11.4 (Intertwiner classification). Let P_ρ be the permutation operator on V induced by φ_ρ . For $\rho \in \text{Sym}(S)$,

$$\text{Hom}_{\varphi_\rho} = kP_\rho, \quad \dim \text{Hom}_{\varphi_\rho} = 1.$$

For the diagonal collapse e_r ,

$$\text{Hom}_{e_r} = U_r \varepsilon, \quad \dim \text{Hom}_{e_r} = 2.$$

Thus the total audited intertwiner dimension is

$$6 \cdot 1 + 3 \cdot 2 = 12.$$

Proof. For an automorphism φ_ρ , the operator P_ρ satisfies

$$P_\rho L_x = L_{\varphi_\rho(x)} P_\rho \quad (x \in M),$$

so $kP_\rho \subseteq \text{Hom}_{\varphi_\rho}$. The Front-C verifier audits the defining linear constraints and gives $\dim \text{Hom}_{\varphi_\rho} = 1$ for the six automorphism spaces; hence equality holds. The formula for Hom_{e_r} and its dimension are Proposition 11.3. Summing over the six automorphisms and three diagonal collapses gives the displayed total. $\square$

12 Binary term functions

A binary term function of RP is a function $M^2 \rightarrow M$ realized by a term in variables x, y . The set of all such functions is denoted $T_2(\text{RP})$.

By the row-owner lemma, every binary term has an owner variable $\text{ow}(f) \in \{x, y\}$. Order the inputs as $(o, n) = (\text{owner}, \text{nonowner})$. On the same-row sector, normalize the common row to 0 and define

$$s_f(i, j) = \text{col}(f(o = (0, i), n = (0, j))).$$

On the cross-row sector, normalize the owner row to 0 and the nonowner row to 1; the output column is independent of the nonowner column, so

$$u_f(i) = \text{col}(f(o = (0, i), n = (1, j))).$$

The opposite cross-row direction is determined by $u_f^-(i) = -u_f(-i)$.

Proposition 12.1 (Binary sector reduction). Every binary term function has a normal signature

$$\Theta(f) = (\text{ow}(f), s_f, u_f).$$

The same-row signature $s_f : S^2 \rightarrow S$ and cross unary signature $u_f : S \rightarrow S$ determine the term function on all row sectors.

Proof. The owner coordinate is Lemma 9.1. Same-row inputs remain in the common input row. Cross-row inputs are governed by the owner row and a unary owner-column signature; the opposite direction is forced by the RP-compatible relabelling exchanging the two nonowner directions. $\square$

The realized same-row signatures form a 21-element set $\mathcal{S}_{21}$. The realized cross unary signatures form a 15-element set $\mathcal{U}_{15}$. The candidate normal forms are

$$\{x, y\} \times \mathcal{S}_{21} \times \mathcal{U}_{15}.$$

Lemma 12.2 (Binary term certificate protocol). The binary term theorem is supported by four finite obligations: T2-SIG exposes $\mathcal{S}_{21}$ and $\mathcal{U}_{15}$; T2-REAL realizes every triple; T2-TRANS checks closure of the two factor transition systems; T2-MULT reconstructs the full 630×630 multiplication table from the owner and factor transitions.

Theorem 12.3. The binary term-function algebra of RP is finite and satisfies

$$T_2(\text{RP}) \cong \{x, y\} \times \mathcal{S}_{21} \times \mathcal{U}_{15}.$$

Consequently

$$|T_2(\text{RP})| = 2 \cdot 21 \cdot 15 = 630.$$

Proof. Proposition 12.1 gives injectivity of the normal signature. The realization certificate supplies one realizing term for every triple, and the finite transition tables reconstruct the product law. Hence the cardinality is $2 \cdot 21 \cdot 15$. $\square$

This is the binary layer of the term algebra. The all-arity clone is left for the open-problem list.

13 Matrix carrier and row-contraction

Under the carrier identification

$$k[M] \cong \text{Mat}_3(k), \quad e_{(r,c)} \leftrightarrow E_{rc},$$

the finite RP multiplication and ordinary matrix multiplication become two bilinear structures on the same nine-dimensional vector-space carrier. The row-contraction bridge records the cross-row branch of RP on matrix units.

Proposition 13.1. Under $k[M] \cong \text{Mat}_3(k)$, the row-contraction extension of the cross-row RP rule is

$$\mu_{\text{cross}}(A, B) = AJ_3B^\top = (A\mathbf{1})(B\mathbf{1})^\top.$$

On matrix units with cross rows,

$$E_{ij}J_3E_{kl}^\top = E_{ik}.$$

Proof. Since $J_3 = \mathbf{11}^\top$,

$$E_{ij}J_3E_{kl}^\top = (e_ie_j^\top)(\mathbf{11}^\top)(e_ke_l^\top) = E_{ik}.$$

For $i \neq k$, this is exactly the cross-row RP rule $(i, j) \cdot (k, l) = (i, k)$. The L3/ matrix-bridge audit tables record the basis-level bridge. $\square$

Thus the matrix bridge expresses the same column-blindness selected by $\mathcal{H}_{acc} = 0$: cross-row multiplication factors through row sums. The same-row branch remains the Steiner/diagonal row-fiber rule.

14 Hidden-continuation operator bridge

The local hidden-continuation identity of Section 4 has an operator interpretation in the left-regular envelope. For a normalized local index q , D_C compares the two stepwise continuation signatures, while D_L compares the two collapsed left translations.

Theorem 14.1 (Hidden-continuation operator identity). Let $g \in \Omega'$ be a completed normal-form magma and let $q = (b, t, a, e, f) \in S^5$. Define

$$xy, \quad yz, \quad z_R, \quad ep_L, \quad ep_R$$

from g as in Theorem 4.1. Then

$$\frac{h_q(g)}{2} = D_C(t, xy, f; b, a, z_R) - D_L(ep_L, ep_R).$$

For brevity write

$$D_C(q; g) = D_C(t, xy, f; b, a, z_R), \quad D_L(q; g) = D_L(ep_L, ep_R).$$

After selection, take $g = \text{RP}$ and write the normalized triple as

$$X = (0, a), \quad Y = (b, e), \quad Z = (t, f).$$

Then

$$XY = (0, xy), \quad YZ = (b, z_R),$$

and

$$(XY)Z = (0, ep_L), \quad X(YZ) = (0, ep_R).$$

With the convention $AB = A \circ B$, the continuation distance $D_C(q; \text{RP})$ is the Hamming-distance contrast between the normalized signatures of

$$L_{XY}L_Z \quad \text{and} \quad L_XL_{YZ},$$

whereas $D_L(q; \text{RP})$ is the Hamming-distance contrast between

$$L_{(XY)Z} \quad \text{and} \quad L_{X(YZ)}.$$

As maps, all four operators lie in $\mathcal{W}_{\text{row}(X)}$; after linearization they lie in $W_{\text{row}(X)}$.

Define

$$I_{\text{tot}}(g) = 6 \sum_{q \in S^5} D_C(q; g), \quad B_{\text{tot}}(g) = 6 \sum_{q \in S^5} D_L(q; g).$$

Thus $H_{\text{tot}}(g) = I_{\text{tot}}(g) - B_{\text{tot}}(g)$. For RP ,

$$I_{\text{tot}} = 8904, \quad B_{\text{tot}} = 1884, \quad H_{\text{tot}} = 7020, \quad N_- = 0.$$

The finalist RC has the same pure hidden-continuation profile.

Proof. The scalar identity is exactly Theorem 4.1 divided by 2: D_C is the sum of the nine continuation disagreement indicators, D_L is the sum of the nine collapsed left-translation disagreement indicators, and h_q is twice their signed difference.

For $g = \text{RP}$, the displayed normalized triple identifies the two continuation signatures with the signatures of $L_{XY}L_Z$ and L_XL_{YZ} , under the convention $AB = A \circ B$. Likewise the collapsed signatures are the left translations by the endpoints $(XY)Z$ and $X(YZ)$. The row-target assertion follows from the composition-envelope law $W_r W_s \subseteq W_r$. The L3/ hidden-continuation bridge audit records the RP/RC comparison and verifies the displayed values. The global frontier status is Theorem 4.2. $\square$

15 Carrier-level compact companions

The carrier identification

$$k[M] \cong \text{Mat}_3(k), \quad e_{(r,c)} \leftrightarrow E_{rc},$$

places the selected finite magma on the same nine-dimensional carrier as the ordinary matrix algebra. A *carrier-level compact companion* is a compact Lie group attached to a specified linear structure on this shared carrier. The ordinary matrix and Lie structures used here are standard [18, 12, 17].

Over fields of characteristic different from 3, ordinary matrix multiplication gives

$$\mathfrak{gl}(3, k) = kI \oplus \mathfrak{sl}(3, k), \quad \text{Mat}_3(k) = kI \oplus \mathfrak{sl}(3, k)$$

as vector spaces. After complex compact form, this is the usual central $\mathfrak{u}(1)$ plus $\mathfrak{su}(3)$ picture.

Over $\mathbb{R}$, the compact three-dimensional branch

$$\mathfrak{so}(3) = \{X \in \mathfrak{sl}(3, \mathbb{R}) : X^\top = -X\}$$

is stable under simultaneous permutation conjugation. A split comparator is $\mathfrak{sl}(V_{std}) \subset \mathfrak{sl}(3, \mathbb{R})$, where

$$V_{std} = \{v : v_0 + v_1 + v_2 = 0\}.$$

Compactness selects the $\mathfrak{so}(3)$ branch among these three-dimensional $\text{Sym}(S)$ -stable carrier branches.

Theorem 15.1. In the real/complex compact-form setting, the carrier identification and the RP-compatible $\text{Sym}(S)$ -action expose the carrier-level compact companions

$$SU(3), \quad SU(2), \quad U(1).$$

Here $SU(3)$ is attached to the ordinary $\mathfrak{sl}(3)$ matrix package, $SU(2)$ is the simply connected compact companion of the selected $\mathfrak{so}(3)$ branch, and $U(1)$ is attached to the central scalar line.

Proof. The matrix-unit carrier gives Mat_3 , hence the ordinary $\mathfrak{sl}(3)$ matrix package and its compact form $SU(3)$. The $\text{Sym}(S)$ -stable compact branch $\mathfrak{so}(3)$ gives $SO(3)$, whose simply connected compact cover is $SU(2)$. The central scalar line gives the compact circle $U(1)$. The L3/ companion-structure bridge audit records these carrier-level compact types. $\square$

16 Signed symplectic lift

This bridge uses the finite pure C/J directed-edge geometry selected in Part II. The continuous statement is a signed projective symplectic realization of that six-edge geometry and completes the carrier-level bridge part of the passport [22].

Let $V = \mathbb{R}^6$ have ordered basis

$$(e_{01}, e_{12}, e_{20}, e_{10}, e_{02}, e_{21})$$

and symplectic matrix

$$\Omega = \begin{pmatrix} 0 & I_3 \\ -I_3 & 0 \end{pmatrix}.$$

Set

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

$$\widehat{C} = \begin{pmatrix} P & 0 \\ 0 & P \end{pmatrix}, \quad \widehat{J}_0 = \begin{pmatrix} 0 & R \\ R & 0 \end{pmatrix}, \quad \widehat{J}_s = \begin{pmatrix} 0 & R \\ -R & 0 \end{pmatrix}.$$

The unsigned $\widehat{J}_0$ induces edge reversal but is anti-symplectic; the signed $\widehat{J}_s$ induces the same projective edge reversal and is symplectic.

Theorem 16.1. The finite pure $\{C, J\}$ directed-edge geometry admits the signed projective dihedral lift displayed above. The unsigned reversal is anti-symplectic, while the signed reversal is symplectic. The finite tick word

$$C, J, C, C, J$$

visits all six directed edges and has a signed symplectic realization.

Proof. Direct multiplication gives

$$\widehat{C}^T \Omega \widehat{C} = \Omega, \quad \widehat{J}_0^T \Omega \widehat{J}_0 = -\Omega, \quad \widehat{J}_s^T \Omega \widehat{J}_s = \Omega,$$

and

$$\widehat{C}^3 = I, \quad \widehat{J}_s^2 = -I, \quad \widehat{J}_s \widehat{C} \widehat{J}_s^{-1} = \widehat{C}^{-1}.$$

Hence the projective classes satisfy the dihedral relations in $\mathrm{PSp}(V, \omega)$. On coordinate lines, $\widehat{C}$ acts as $(01 \ 12 \ 20)(10 \ 02 \ 21)$, and $\widehat{J}_s$ acts as edge reversal. Starting from 01, the tick word gives

$$01 \mapsto 12 \mapsto 21 \mapsto 10 \mapsto 02 \mapsto 20.$$

The signed-lift matrices and tick-word audit are recorded in the L3/ signed-symplectic bridge tables. $\square$

The signed lift completes the post-selection passport of RP.

17 Conclusion

The normal-form landscape

$$\Omega' \cong S^{21}$$

is mapped by the associativity atlas and the $\mathcal{H}$ -frontier. The four internal selector predicates

$$\mathcal{H}_{acc} = 0, \quad H_{diag} = 0, \quad (r, r)^2 = (r, r), \quad \text{AbsTrans} = \{C, J\}$$

select the right-row projection magma RP. After this selection, Part III records the finite post-selection passport: operator envelope, term algebra, endomorphism/intertwiner classification, row-contraction matrix bridge, hidden-continuation operator identity, carrier-level compact companions, and signed symplectic lift.

The manuscript therefore has the publication chain

$$\begin{array}{ccc} \Omega' \cong S^{21} & \xrightarrow{\text{selector}} & \text{RP}, \\ \text{RP} & \xrightarrow{\text{linearization}} & \text{finite operator, term,} \\ & & \text{and carrier passport} \end{array}$$

Companion material

A Finite certificate protocol

The finite support package is available in the public GitHub repository:

<https://github.com/burundai-t/finite-row-graded-magmas-over-f3>.

The finite algebraic language follows the standard binary-system, universal-algebra, design, quasigroup, and clone-theory sources [5, 6, 7, 8, 11, 9, 23, 24, 25, 27, 20]. The certificate style is the standard finite-proof pattern: a human-readable reduction is paired with reproducible finite artifacts [1, 2, 13, 15, 16, 21].

component	role	manuscript interface
L1/	finite landscape and associativity reference checks	normal form, associativity atlas, end-point loci, partition function
L2/	compact selection and pure C/J survivor checks	selector predicates, directed-edge absorption, uniqueness theorem
L3/	selected-RP operator and bridge checks	operator envelope, T_2 , matrix bridge, compact companions, signed symplectic lift
<code>mathcal_H/</code>	global $\mathcal{H}$ -frontier bundle	local distance reduction, H1–H4 frontier obligations
repository interface	<code>README.md</code> ; <code>SUPPORT_INDEX.txt</code>	reviewer commands and file-by-file index

The compact table families are:

family	location	purpose
L1 landscape atlas	<code>L1/tables/</code>	associativity distribution, endpoints, orbit representatives, controlled $\mathcal{H}$ comparison
L2 selection	<code>L2/tables/</code>	access-zero, diagonal compression, pure C/J survivor separation
L3 linearization pass-port	<code>L3/tables/</code>	positive composition envelope, binary term algebra, matrix bridge, signed symplectic lift, fiber algebra
$\mathcal{H}$ frontier	<code>mathcal_H/</code>	H1 unrestricted range, H2 pure-frontier UNSAT, H3 witnesses, H4 spike classifier

A.1 Global $\mathcal{H}$ -frontier certificate handles

The global $\mathcal{H}$ -frontier theorem is checked by a separate `mathcal_H/` bundle. At manuscript level the implication is

$$\text{local distance reduction} + \text{H1–H4 certificate obligations} \implies \text{global frontier theorem}.$$

handle	bundle location	theorem component
HF-RED, HF-CERT-IF	bundle theorem interface	local distance reduction and local signature formulas
H1	h1/	unrestricted full-landscape range $rawH \in [-378, 1217]$
H2	h2/	pure-frontier exclusion at $rawH \geq 1171$
H3	embedded witness verifier	RP and RC at $rawH = 1170$, $N_- = 0$
H4	h4/	twelve-point signed-cancellation spike at $rawH = 1217$

The H2 obligation is the finite UNSAT statement

$$(h_q \geq 0 \ \forall q \in S^5) \wedge rawH \geq 1171$$

on the depth-9 live frontier:

$$18540/18540 \text{ UNSAT}, \quad \text{SAT} = 0, \quad \text{UNKNOWN} = 0.$$

The light static coverage verifier for the accepted H2 raw artifacts is:

```
cd mathcal_H/h2
PYTHONDONTWRITEBYTECODE=1 python3 verify_h2_coverage_light.py
```

A.2 Dependency summary

result family	support role
normal form and selector proof	hand finite arguments with L2 anchor checks
associativity atlas	block reduction plus L1 endpoint and partition-function tables
global $\mathcal{H}$ -frontier	local distance reduction plus H1–H4 certificate bundle
operator envelope and row-fiber algebra	analytic row-owner reductions plus L3 multiplication tables
endomorphisms and intertwiners	finite row-fiber classification with linear intertwiner audit
binary term algebra	sector reduction plus realization and transition certificates
matrix, compact-companion, and symplectic bridges	carrier-level calculations supported by L3 bridge audits

B One-page passport of RP

The selected point is

$$\text{RP} = (a, b, d) = (1, 2, 000).$$

It has the finite economy and compression profile used by the selector:

$$\mathcal{H}_{acc} = 0, \quad H_{diag} = 0, \quad (r, r)^2 = (r, r), \quad a \neq b, \quad d = 000.$$

Thus its cross-row rule is column-blind, its same-point diagonal is compressed and anchored, and its two cross-row anchors are nondegenerate. The Landauer/Shannon provenance of the first two predicates is zero hidden-column access and zero diagonal entropy [19, 3, 4, 26, 10].

Proposition B.1 (Nondegenerate anchors). Among the nine column-blind rules $h_{a,b}$, the condition $a \neq b$ leaves exactly six rules.

Proof. There are nine pairs $(a,b) \in S^2$, and the three pairs (a,a) are excluded. $\square$

Proposition B.2 (Local associativity finalists). With $d = 000$, the six nondegenerate column-blind rules have

$$\text{Assoc}_{000} = 219$$

exactly at $(a,b) = (1,2)$ and $(2,1)$, namely RP and RC. The other four nondegenerate rules have $\text{Assoc}_{000} = 273$.

Proof. By Proposition 3.3, on the column-blind, $d = 000$ stratum,

$$\text{Assoc}_{000}(h_{a,b}) = 3(73 + 18[a = 0] + 18[b = 0] + 54[a = b]).$$

On $a \neq b$, this is minimized exactly when neither anchor is 0, giving $(1,2)$ and $(2,1)$. $\square$

The selected directed-edge feature is pure head-row absorption.

Lemma B.3 (Absorption passport). For RP,

$$\text{AbsTrans}_{\text{RP}} = \{C, J\}.$$

The row-complement comparison point $\text{RC} = (a, b, d) = (2, 1, 000)$ has

$$\text{AbsTrans}_{\text{RC}} = \{C^{-1}, C^{-1}J\}.$$

Proof. Let $x = (r, c) \in M^\times$. For RP, the cross-row rule is $h(r_x, r_y) = r_y$, so absorption is $r_y = c$. The two absorbed neighbours are (c, r) and $(c, \overline{rc})$, namely $J(x)$ and $C(x)$.

For RC, the cross-row rule is $h(r_x, r_y) = \overline{r_x r_y}$. The absorption equation $\overline{r r_y} = c$ gives $r_y = \overline{rc}$. The two absorbed neighbours are $(\overline{rc}, r)$ and $(\overline{rc}, c)$, namely $C^{-1}(x)$ and $C^{-1}J(x)$. $\square$

This one-page passport records RP at the selection boundary: access compression, diagonal compression, anchored self-action, nondegenerate anchors, local associativity finalist value, and the head-row absorption pair $\{C, J\}$. The adjacent row-complement finalist has the drifted absorption pair $\{C^{-1}, C^{-1}J\}$.

C Theoretical-Informatic Reading of uniquely selected RP

The four selector predicates (together with the supporting structural properties of the selected magma RP) admit the following information-theoretic and structural reading:

Mathematical condition	Information-theoretic / structural reading
$\mathcal{H}_{\text{acc}} = 0$	Do not read hidden columns; avoid Landauer-facing erasure of hidden information.
$(a \neq b)$, for RP $((a, b) = (1, 2))$	Non-triviality. The output depends on the right public direction; interaction does not collapse into an empty response.
$H_{\text{diag}} = 0$	Minimize Shannon entropy on the diagonal — do not distinguish diagonal types where distinction is unnecessary.
min Assoc ₀₀₀ / low local associativity	The order of application has meaning (the sequence of operations carries information).
pure (C/J) and signed symplectic lift	Directed-edge structure is preserved without drift; at the linear level, phase-volume preservation appears.
$((r, c)^2 = (r, r))$	Self-application returns to the element's own row-fixed point.

These six readings can be distilled into two complementary axes of a single principle:

Economize, but do not descend into triviality.

Preserve structure and information — without unnecessary expenditure.

D Open bridges and extensions

The following fronts are subsequent work beyond the theorem package proved in this manuscript.

D.1 Functorial RP– $\mathfrak{gl}(3)$ bridge

The shared carrier

$$k[M] \cong \text{Mat}_3(k)$$

supports the finite RP multiplication and ordinary matrix multiplication. The row-contraction bridge

$$\mu_{\text{cross}}(A, B) = AJ_3B^\top$$

aligns the cross-row branch with matrix rank-one contraction. The next problem is to identify a universal or functorial construction from the finite magma to the matrix/Lie package. The matrix and Lie background used in the carrier-level bridge is standard [18, 12, 17].

D.2 Symplectic development of pure C/J

The pure directed-edge geometry has an explicit signed symplectic lift. The next layer is the associated continuous symplectic geometry: canonical (q, p) -charts, Hamiltonian paths, and coordinate-free separation between head-row reversal and row-complement drift. This is the natural continuation of the finite lift in the language of symplectic geometry [14, 22].

D.3 Unified selection principle

The uniqueness theorem uses four finite predicates: access-zero, diagonal entropy zero, row-anchor idempotence, and pure C/J . A sharper formulation would derive these predicates from one natural functional whose global minimizer on Ω' is the RP representative.

D.4 All-arity term algebra

The binary term algebra satisfies

$$T_2(\text{RP}) \cong \{x, y\} \times \mathcal{S}_{21} \times \mathcal{U}_{15}, \quad |T_2(\text{RP})| = 630.$$

The all-arity clone remains to be classified. The relevant background is the classical clone-theoretic framework [24, 25, 27, 20].

D.5 Larger carriers

Natural extensions include analogues over larger finite fields or cyclic sets and directed-edge C/J geometries beyond the triangle. The standard combinatorial-design and quasigroup references provide the ambient language for these variants [11, 9, 23].

D.6 Relaxing AX2

The present landscape fixes the same-row Steiner rule. The relaxed landscape asks whether that same-row rule can itself be selected intrinsically.

Note on AI assistance

This work was developed with assistance from large language models. The author specified the conceptual framework, constraints, and target properties of the objects under study. Within this framework, AI tools were used for computational exploration, code drafting, and structuring intermediate results. All mathematical definitions, statements, and theorems presented in the paper were established and independently verified by the author. The finite claims are backed by exhaustive checks recorded in the accompanying support package.

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